LESSON 2.1

Understand Statistical Questions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can tell the difference between a statistical question and a non-statistical question.

WHY THIS MATTERS You've just been hired as the data analyst for a youth basketball league. The head coach hands you a list of questions the staff wants to investigate about their players. Some questions will give you interesting data with variety, while others have just one fixed answer. Your job: figure out which questions are worth investigating with data. Two of the coach's questions are: "How tall is our point guard?" and "How tall is each player on the roster?"

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Statistical Question *Spanish: Pregunta estadística*	A question answered with data that are expected to vary.	How tall is each student?
Non-statistical Question *Spanish: Pregunta no estadística*	A question that does not anticipate variability in the data used to answer it. It may ask for one fact or measurement.	How tall is the flagpole?
Data *Spanish: Datos*	Facts and numbers you collect, like answers to a survey.	0, 2, 5, 1 pets
Variability *Spanish: Variabilidad*	How spread out the numbers are.	0, 2, 5 pets — answers vary
Survey *Spanish: Encuesta*	Collecting facts by asking people questions.	Asking classmates one question
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	A distribution is symmetric when its left and right sides have approximately matching shapes around the center.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Which is statistical: “How many siblings does each student have?” or “How many sides does a triangle have?”

1. Sibling counts differ: 0, 1, 3.
2. A triangle always has 3 sides.

Answer: The sibling question is statistical; the triangle question is not.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New question: "How many pets does each student in our class have?" Will every student give the same answer, or different answers?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This question _____ statistical because _____ .
- I expect the answers to _____ from person to person.
- A non-statistical question has _____ answer.
- The data I would collect might look like _____ .

Word bank statistical | non-statistical | variability | data | survey | vary | answers | collect | fixed | different | question | typical

WATCH OUT!

A common mistake

A common mistake in Statistical Questions and Data is judging a question by its topic or its numbers instead of its variability. Students see a question about ONE player, ONE game, or ONE fixed fact

LESSON 2.1

Understand Statistical Questions

Version A · Supported practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each question to its type.

_____	How tall is each swimmer on the team?	A	Statistical
_____	What color is the team mascot?	B	Not Statistical
_____	How many laps does each runner complete?	C	Statistical
_____	How many players are on a soccer team?	D	Not Statistical

HINT Read each question on the left and underline the word 'each' when you see it — it's a clue that many people are being measured.

2 **Write the name of the correct group on each line.**

Sort each question by type.

Groups: Statistical Question | Not a Statistical Question

_____ How many books did each student read this month?

_____ What grade did each student earn on the test?

_____ How many days are in a week?

_____ What is the school mascot?

HINT Statistical questions expect different answers from different people or observations.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which of the following is a statistical question?

(A) How many points did each player score this season?

(B) How many points are scored for a free throw?

(C) What sport does this team play?

(D) How many halves are in a soccer game?

HINT Re-read each question and ask: would different people or games give different answers, or is there just one fixed answer?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which question is NOT a statistical question?

(A) How many home runs did each player hit this year?

(B) How many miles does each runner train per week?

(C) How many innings are in a baseball game?

(D) How tall is each player on the basketball team?

HINT Careful — this asks for the question that is NOT statistical: the one with a single fixed answer instead of varying data.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Why is 'How many hours of sleep did each student get last night?' a statistical question?

(A) Because different students got different amounts of sleep

(B) Because it asks about sleep

(C) Because hours are numbers

(D) Because the answer is always 8 hours

HINT The question asks WHY it's statistical — focus on what happens when many students answer, not on the topic of sleep.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Question** How tall is each player on the basketball team?
2. **Student decision** This is NOT a statistical question.
3. **Student reason** Height is just one number, so every answer will be the same.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT The word "each" is a clue that you need data from many players.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about statistical questions compared to non-statistical questions?

SENTENCE STARTER Statistical questions expect ___ in the answers because ___. Non-statistical questions have only ___ answer because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.1

Understand Statistical Questions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Decide if each question is statistical and explain why.

Question	Statistical?	Why?
How many push-ups can each student do in one minute?		
How many sides does a basketball court have?		
How far does each player kick the ball?		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each question with what makes it statistical or not.

_____ How fast does each sprinter run the 100m?

_____ How long is an Olympic pool?

_____ How many goals did each player score this season?

_____ How many innings are in a baseball game?

(A) Variability — Statistical

(B) One answer — Not Statistical

(C) Variability — Statistical

(D) One answer — Not Statistical

3 **Write the name of the correct group on each line.**

Match each non-statistical question with its statistical rewrite.

Groups: **Original: 'How many students are in 6th grade?'** **Original: 'What time does school start?'**

_____ Rewritten: 'How many hours does each 6th grader study per week?'

_____ Rewritten: 'What time does each student arrive at school?'

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A coach asks: "How many hours does each athlete on the track team sleep the night before a meet?" Why is this a statistical question?

(A) Because it is about sports

(B) Because it has a numerical answer

(C) Because the answers will vary from athlete to athlete

(D) Because the coach asked it

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which of these is a statistical question?

(A) What time is it right now?

(B) What is my shoe size?

(C) What is 7 × 8?

(D) How tall are the students in my class?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Question** How many points did the team score in its last completed game?
2. **Student decision** This IS a statistical question.
3. **Student reason** It is about basketball and the answer is a number, so it must be statistical.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about statistical questions compared to non-statistical questions?

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LESSON 2.1

Understand Statistical Questions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Rewrite each non-statistical question to make it statistical and explain the variability.

Non-Statistical Question	Rewritten as Statistical	What Variability Exists?
How many players are on the team?		
What sport does this league play?		

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A survey asks 30 students how many books they read last month, collecting 30 answers. What makes this a statistical question?

(A) It is about books

(B) Everyone reads the same number

(C) The answers vary from student to student

(D) It has exactly one correct answer

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which rewrite turns 'What is my resting heart rate?' into a statistical question?

(A) Is my heart rate high?

(B) What is a heart rate?

(C) What are the resting heart rates of students in my class?

(D) What is the teacher's heart rate?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Find the mistake. Explain it, then show the correct work.**

Find the Reasoning Error

1. **Question** How many touchdowns did the quarterback throw in each game this season?
2. **Student says** This is NOT a statistical question.
3. **Reason** It only asks about one player, so there is no variability.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 **Answer in complete sentences.**

Write one statistical question and one non-statistical question about your school's cafeteria. Explain why each question is or is not statistical, and describe what the data would look like for the statistical question.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What pattern do you notice about statistical questions compared to non-statistical questions?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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